EduShiksha

v1.0 · 20 October 2026

ROADMAP · PLAYBOOK · CHECKLIST

Roadmap aur School Management Playbook

Ab tak kya bana, aage kya aur kaise banana hai, aur ek school chalane ke liye kya-kya cover karna hoga

KISKE LIYE

Anuj (project owner)

BASIS (KIS PAR BASED)

Repo ka state (branch anuj-dev, 172 backend tests), docs/01 se 05 (BRD, FRD, journeys, edge cases), EasyEdulab aur Infinity Medisetu ke screenshots, aur India ke school rules par web research.

BHASHA

Hinglish: Hindi (Roman) + English technical words

Is document ko kaise padhein

Ye document isliye hai ki bina kisi aur se review karaye bhi aap poora school management cover kar sakein: kya banana hai, kis order mein, kaise, kya dhyan rakhna hai aur kya nahi banana. Har point chhota aur seedha rakha hai.

Part 1	Abhi kahan hain: kya ban chuka, kya adhoora, kya jaan-boojh ke baad ke liye chhoda.
Part 2	Golden rules: jo har module mein lagti hain (data isolation, Academic Year, paisa, audit).
Part 3	Roadmap: kis order mein banayein aur kyun. Pilot school ke liye minimum kya chahiye.
Part 4	Module playbooks: har module ke screens, features, data, rules, edge cases aur 'done kab'.
Part 5	Process: ek module ko kaise banayein, test karein, release karein.
Part 6	India ke rules (DPDP, APAAR, UDISE+, RTE, CBSE, DLT, tax) aur product mein inka matlab.
Part 7	School ka saal: mahine-dar-mahine school kya karta hai aur system ko kya sambhalna hai.
Part 8	Decisions jo aapko lene hain (mere recommendation ke saath).
Part 9	Go-live checklist aur agle kaam.
Appendix	Kaunsa data saal se bandha hai, glossary, sources.

Teen cheezein yaad rakhne ki: (1) Har record **school (tenantId)** se bandha. (2) Jo cheez saal ke saath badalti hai wo **Academic Year** se bandhi. (3) Paisa, marks, attendance **kabhi delete nahi**, sirf reversal + audit.

Dhyan: Part 6 ke rules (DPDP, APAAR, RTE, tax) web research aur secondary sources par based hain. Ye **legal advice nahi hai**: har point par "Verify" tag dekho aur asli sarkari text / vakeel / CA se confirm karo.

PART 1 Abhi kahan hain

Foundation taiyar hai, school ke roz ke kaam abhi shuru nahi hue. Login, school ko onboard karna, approve karna, data isolation aur Classes & Sections bane hain. Students, fees, attendance, exams, notices: ye sab abhi 0%.

Ban chuka hai

- Super admin: login / logout, 15 minute ka access token + 30 din ka rotating refresh token, forgot / reset password (email se)
- School invite (email link, 72 ghante) aur login page se self sign-up
- 4-step onboarding (school, address, aap aur classes, review), server par save, validations
- Approval flow: approve / reason ke saath wapas bhejo / dobara submit; suspend aur reactivate
- Super admin dashboard, Schools list, review page
- **Data isolation**: `tenantScope` plugin, 'har model scoped ya listed' registry test, har module ke liye `itKeepsSchoolsApart` test helper
- Classes, Sections, Departments: approval par onboarding ki range se classes khud ban jaati hain, quick setup, ek-click section, re-order, bina section wali class
- React frontend (Vite, RTK Query), design system, tooltips, har feature ka README
- Automated tests: 172 backend tests pass

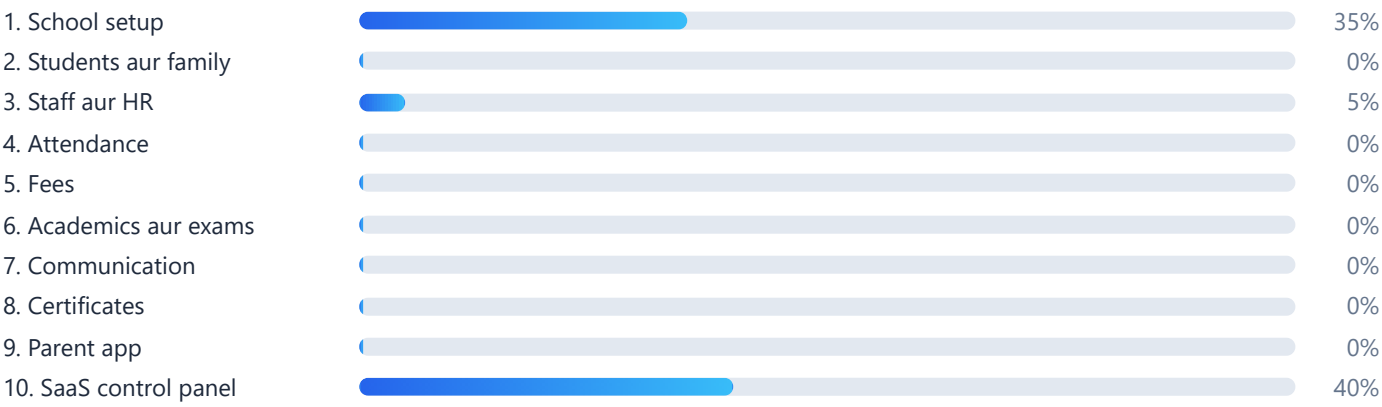
Adhoora

Area	Status
School setup (M1)	Classes done. Academic year, subjects, branches, logo / letterhead baaki.
SaaS control (M10)	Approve / suspend done. Plans, billing, module toggles, analytics, tickets baaki.
Roles	7 roles code mein hain, par staff ko invite karne ka flow (Step 5) abhi nahi.

Abhi shuru nahi hua

Students aur Admission · Parents / Family · Staff directory aur HR · Attendance · Fees · Timetable · Homework · Exams aur Report cards · Notices / Messaging · Certificates aur ID cards · Parent / Student app · Reports aur UDISE+ export · Subjects · Academic Year

BRD ke 10 modules mein progress (andaza)



Ye andaza hai (effort ke hisaab se), exact nahi. Poore Phase 1 ka lagbhag 8-10% bana hai: foundation ke liye zaroori, par product abhi school ke daily kaam ke layak nahi.

Jaan-boojh ke baad ke liye chhoda gaya

Ye aapke hi faisle the (pehle product, phir hardening). Neeche "kab" column batata hai ki kab karna zaroori ho jata hai.

Cheez	Kab karna hai	Source
Change password aur 'sab jagah se logout'	**Pilot se pehle**	Parked list + FRD security
Audit log (approvals, fees, marks, attendance corrections)	**Fees se pehle**	FRD auditability
Native bcrypt / argon2 (abhi 8 parallel login ~1.5 s event loop rok dete hain)	**Launch se pehle (sabse upar)**	Parked list
Email uniqueness: parent / teacher do schools mein	**Students module se pehle**	Part 8, D1
Onboarding: UDISE required + unique, school-name uniqueness change, Terms / Privacy checkbox, logo / documents	**Pilot se pehle**	Onboarding gaps
Multi-branch model	**Students se pehle**	Part 8, D2
HttpOnly refresh cookie + CSP	**Launch se pehle**	Parked list
MFA (TOTP) super admin ke liye; .env se SUPER_ADMIN_PASSWORD hatao	**Launch se pehle**	Parked list
Redis par rate limits + sahi `trust proxy` (school ek NAT ke peeche 600 req / 15 min cross kar sakta hai)	**Launch se pehle**	Parked list
Mongo transactions (replica set) accept-invite jaise multi-step kaam ke liye	**Launch se pehle**	Parked list
Cancel / fix an invited school, school ke contact email ka change	**Jab zaroorat pade**	Onboarding gaps

PART 2 Golden rules

Ye 12 rules har module par lagti hain. Inme se kuch (data isolation) code mein enforce ho chuki hain, baaki aap aur main review mein lagayenge.

1 **Har record school (tenant) se bandha**
Har school-wale model par `tenantScope` plugin, har service function mein `tenantId` pehla argument, aur har module ka `<module>.isolation.test.ts` (`itKeepsSchoolsApart`). 'Doosre school ko 404' ek test hai, vaada nahi.

2 **Jo cheez saal ke saath badalti hai wo Academic Year se judti hai**
Enrolment, fee structure / ledger, attendance, exams, timetable, class teacher, homework: sab mein `academicYearId`. Classes, sections, departments, subjects, student ka insaan wala record, staff ki pehchaan **year-free** rehte hain (Class 5 har saal wahi class hai). Full list Appendix A mein.

3 **Paisa, marks, attendance kabhi hard-delete nahi**
Galti hui to **reversal / correction entry** + reason + audit log. Purana record dikhta rahe. Receipts cancel hote hain, mitaaye nahi jaate.

4 **Paisa integer (paise) mein**
Float kabhi nahi. Receipt number atomic counter se, financial year ke saath, kabhi repeat nahi. Gateway webhook idempotent.

5 **Role ka check backend mein**
`requireRole` asli rok hai; UI sirf button chhupata hai. Teacher sirf apni classes dekhe.

6 **Sensitive data kam, masked, aur logged**
Aadhaar / PAN / bachchon ke phone: sirf zaroori ho to lo, mask karke dikhao, role se gate karo, har view audit mein. Logs / exports mein default se nahi.

7 **Policy code mein nahi, school ki setting mein**
'TC se pehle dues clear?', 'dues par report card roke?', 'late fee kitni?' jaise sawal har school alag tay karta hai.

8 **Ek page, ek baar, poora**
Backend + tests + UI + real-browser check + README, phir agla page. Adhoora module 'baad mein' nahi.

9 **Copy sach ho**
Jo bana nahi use menu mein 'Soon' ya ghalat claim se mat dikhao (aapne pehle hi sab 'Soon' items hatwaye).

10 **Mobile-first jahan roz ka kaam ho**
Attendance, homework, fee payment phone par 1 minute mein hone chahiye.

11 **Import / export har jagah**
Schools Excel se aate hain aur Excel / PDF maangte hain: students, staff, fees ka template-based import aur har list ka export.

12 **Jo jaan-boojh ke chhoda wo likha ho**
Parked list (Part 1) aur Decisions (Part 8) ko zinda rakho; naya module banane se pehle dekho.

PART 3 Roadmap: kis order mein aur kyun

Order dependency se tay hota hai: jo cheez dusre ko chahiye wo pehle. Neeche ek lambi chain hai; beech ke kai kaam saath-saath bhi ho sakte hain.

Decisions → Audit + password → Academic Year → Staff + roles → Subjects → Allocation → Students → Attendance → Notices → Fees → Timetable → Exams + Report cards → Certificates → Parent portal

#	Step	Effort	Kyun is jagah	Kya chahiye	Done jab
0	**Do decisions (code nahi)**	S	Parent identity aur branch model tay nahi to Students ka data shape galat ban sakta hai.	-	Part 8 ke D1 aur D2 ka jawab docs mein likha hai.
1	**Platform base: audit log, change password, logout-all**	M	Fees aur marks se pehle audit chahiye; baad mein jodna mushkil aur khatarnak.	-	Audit log sab sensitive actions par, tests.
2	**Academic Year**	M	Baaki sab isse judte hain. Aapka naya rule yahin lagta hai.	-	Saal banta hai, switch hota hai, rollover preview.
3	**Staff invite, roles, permissions (Step 5)**	L	Admin akela sab nahi karega; teacher ko login chahiye.	D1	Invite se login tak, role-wise menu.
4	**Subjects + class-subject map**	S	Timetable, marks, homework ka base.	2	Subjects banen aur classes se judein.
5	**Teacher allocation + class teacher**	M	EasyEdulab ka 'Class Teacher' column; teacher ko sirf apni classes.	3, 4	Allocation aur class teacher dikhe.
6	**Students, Enrolment, Parents, Documents**	XL	School ka dil. Student aur Enrolment alag rakhne se year rule apne-aap lagta hai.	2, D1, D2	Admission se profile tak, import, promotion dry-run.
7	**Excel import (students, staff)**	M	Pilot school ka purana data aise hi aayega.	6	Template, validation report, dry-run.
8	**Attendance + holiday calendar**	M	Roz ki aadat; chhota; adoption badhata hai.	5, 6	Ek class 1 minute mein; lock aur audit.
9	**Notices (in-app) + messaging base**	M	Attendance alert aur fee reminder ka base. DLT / WhatsApp paperwork saath-saath shuru karo.	6	Notice sahi audience ko, read log.
10	**Fees: heads → structure → collection → receipt → dues → concession → reminders**	XL	Sabse bada fayda (BRD BO-01, BO-02) aur sabse zyada risk: tests sabse zyada.	6, 1, 9	Ledger = receipts; daily report; bounce, part payment.
11	**Timetable + substitution**	L	Teacher productivity; leave se judta hai.	4, 5	Conflict block, teacher ka aaj ka schedule.
12	**Homework / classwork**	M	Parent app ko roz kholne ki wajah.	4, 5, 9	Publish, notification.
13	**Exams, marks, report cards**	XL	Result ka din school ke liye sabse tension ka din hai.	4, 6, 8	Poori exam cycle, snapshot, publish approval.
14	**Certificates (TC, Bonafide) + ID cards**	M	Roz ka kaam; legal.	6, 10	Serial never repeats; register.

#	Step	Effort	Kyun is jagah	Kya chahiye	Done jab
15	**Parent / Student portal (web pehle)**	XL	Parent ke liye asli value ek jagah.	6, 8, 10, 9	Sibling switcher, attendance, fees, notices.
16	**Online payments + WhatsApp / SMS integration**	L	Gateway aur DLT / WhatsApp approvals mein samay lagta hai.	10, 9	Webhook reconcile; fallback SMS.
17	**Reports, analytics, UDISE+ export**	M-L	Principal aur trust ko numbers; saal ke end ka sarkari kaam.	6 se 13	Numbers source se match.
18	**HR leave, Transport, Library, Hostel, Assets**	M each	Pilot ke baad school ki maang se chuno.	6	Maang confirm hone par.
19	**SaaS: plans, billing, module toggles, analytics, tickets**	L	Business ke liye; pilot ke baad.	Pilot	Plan badalne par features on / off.

Effort: S = 1-3 din, M = 1 hafta, L = 2 hafte, XL = 3+ hafte (ek focused developer ka andaza; Classes & Sections jaisa page ~ M). Ye sirf tulna ke liye hai.

> **Attendance pehle ya Fees pehle?** Mera recommendation: Students ke baad **Attendance** (chhota, roz ka, teacher ki aadat banata hai), phir **Fees** (sabse bada aur risky). Agar pilot school ka asli dard fees hai to dono ki jagah badal sakte hain: dependency dono ki Students hi hai.

Pilot school ke liye minimum (go-live cut)

Ek school ko 'asli use' ke liye minimum ye chahiye. Isse pehle pilot mat shuru karo, isse zyada bhi pehle mat banao:

Pilot mein chahiye

- ☐ Academic Year, Staff invite + roles, Subjects, Teacher allocation
- ☐ Students + Enrolment + Parents + Excel import
- ☐ Attendance (parent ko absent ka SMS ke saath)
- ☐ Fees: counter collection + receipts + dues report + cheque / part payment (online baad mein)
- ☐ Notices (in-app + SMS)
- ☐ Exams + marks + ek board ka report card format (pilot school wala)
- ☐ TC aur Bonafide
- ☐ Audit log, backups, change password, Terms / Privacy

Baad mein

- Timetable auto-generate, online exams, GPS transport, payroll, library, hostel
- Native mobile app (pehle responsive web), custom report builder
- Multi-branch trust dashboard (decision D2 ke hisaab se)

PART 4 Module playbooks

Har module ka ek card: kyun chahiye, kaun use karega, screens, features (MUST pehle, SHOULD phir, LATER baad mein), data, rules, edge cases aur 'done kab'. Order roadmap ke hisaab se hai.

> **Edge cases ke code:** S-x.y jaise tags aapke `docs/04_Discussion_Scenarios_and_Edge_Cases.md` ke scenarios hain: pilot school ke saath unhe 'Confirmed / Needs change' mark karna UAT ban jayega.

M1 Cross-cutting Platform (har module ke neeche)

Roadmap step 1 | MVP | Effort Ongoing

Ye features alag page nahi, par inke bina koi module 'production' nahi: audit, import / export, files, jobs, backup.

Kaun use karega: Poora system

Screens

- Settings > School profile (logo, letterhead, policies)
- Audit log viewer (Admin)
- Import / Export center
- Change password / logout everywhere

Features

MUST **Audit log** (immutable): fee entries, marks, attendance correction, TC, Aadhaar view; kisne, kab, purana to naya

MUST Change password aur 'sab jagah se logout'; MFA super admin ke liye (parked list)

MUST Excel import framework: template download, validation report, dry-run, rollback

MUST File storage service (S3-compatible, India region), signed URLs

MUST Background jobs (queue): reminders, reports, notifications; Redis

MUST Daily backups aur **restore ka test** (RTO / RPO likho)

SHOULD Monitoring: errors (Sentry), uptime, slow-query alerts

SHOULD Hindi UI (i18n), feature flags, policy settings (TC dues block, report card withhold)

Data

- `AuditLog { tenantId, actorId, action, entity, before, after, at }` (append-only)

Rules

- Audit mein password / token kabhi nahi
- Policies code mein hard-code nahi, school-wise setting

Edge cases (zaroor socho)

- Bade import mein aadha fail
- Backup se restore ke baad sequence / receipt counters

Kya chahiye pehle: Pehle audit log + import framework banao, phir Students.

Done jab:

- [ ] Fees se pehle audit log live
- [ ] Restore ka ek real test ho chuka ho

M2 Academic Year & Terms

Roadmap step 2 | MVP | Effort M

Har student, fee, attendance aur exam record kisi saal se bandhta hai. Saal badalne par purana data bina bigde safe rehna chahiye. Ye module sabse pehle banta hai kyunki baaki sab isse judte hain.

Kaun use karega: School Admin | Principal

Screens

- Settings > Academic Year: saalon ki list (current wala marked)
- Naya saal banao: naam (2026-27), start, end, terms (Term 1 / Term 2)
- Header mein 'working year' selector (purana saal sirf dekhne ke liye)
- Rollover wizard: naya saal shuru karne ka step-by-step

Features

MUST Ek school ka ek hi **current** saal; onboarding ke `sessionStartMonth` se default dates (April-March)

MUST Saal ke andar terms/semesters (exam aur fee installments inhi par chalenge)

MUST Purana saal select karke dekho: sab kuch read-only

MUST Rollover wizard: structure copy (class-subject map, fee structure), students ka bulk promotion, purane dues ka carry-forward

SHOULD Closed saal ko lock karo; Admin reason likh kar reopen kare (audit log ke saath)

LATER Alag-alag branch ke alag calendars

Data

- `AcademicYear { tenantId, name, startDate, endDate, terms[], status: upcoming | current | closed }`
- Har saal-se-bandha record mein `academicYearId` (neeche Appendix A ki table dekho)

Rules

- Do saal overlap nahi kar sakte; current exactly ek
- Jis saal mein data hai use delete nahi kar sakte, sirf close
- Year-free models ki list ek registry test mein hogi (jaise tenantScope ka `PLATFORM_MODELS` test): naya model bina year ke ban hi na sake

Edge cases (zaroor socho)

- School ka session Jan / Apr / Jul se shuru hota hai
- Aakhri mahine mein hua admission kis saal mein jayega
- Rollover ke din purane dues, detained (fail) students, TC wale students
- Saal beech mein class split (5-A se 5-A aur 5-B): attendance/fee history na toote (S-1.3)

Kya chahiye pehle: Kuch nahi (Classes pehle se bane hain).

Done jab:

- ☐ Saal banta/edit hota hai, switch hota hai, rollover ka preview dikhta hai
- ☐ Isolation test + 'year-free registry' test pass
- ☐ README mein 'kaunsa model year-bound hai, kaunsa free' likha hai

M3 Staff Invite, Roles & Permissions (Step 5)

Roadmap step 3 | MVP | Effort L

School admin akela sab kaam nahi karega: principal, teacher, accountant ko bulana hai aur har ko sirf uska kaam dikhana hai. README ka agla step yahi hai.

Kaun use karega: School Admin | Principal

Screens

- Staff > Invite staff (email + role), Pending invites (resend / cancel)
- Staff list (search, filter by role, active / inactive)
- Role permission matrix (dekhne ke liye; FRD section 3 jaisa)
- Apna profile + password change

Features

MUST Email se invite, one-time link (school invite jaisa hi token flow), link se password set

MUST Roles: principal, teacher, accountant (baad mein receptionist, librarian...)

MUST Permission backend mein enforce (`requireRole`); UI sirf chhupata hai

MUST User ko deactivate karo: sessions turant band, data safe (suspend wala logic pehle se hai)

SHOULD Admin 'view as teacher' dekh sake ki teacher ko kya dikhta hai

SHOULD Role ke andar chhote overrides (jaise teacher notice bhej sake ya nahi)

LATER Custom roles, SSO

Data

- `User { role, tenantId, isActive }` + invite token
- Staff profile alag module (HR) mein

Rules

- Teacher sirf apni classes/sections dekhega (allocation se)
- Principal = Admin minus billing/subscription
- Email abhi poore platform mein unique hai: **decision pending** (Part 8)

Edge cases (zaroor socho)

- Ek teacher do schools mein padhata hai
- Principal badal gaya / admin ownership transfer
- Invite expire ho gaya, galat email par gaya

Kya chahiye pehle: Email/identity ka decision.

Done jab:

- [ ] Invite → accept → login → role ke hisaab ka menu, end-to-end
- [ ] Har role x important routes ka permission test
- [ ] `<module>.isolation.test.ts` pass

M4 Subjects & Syllabus

Roadmap step 4 | MVP | Effort S

Timetable, marks, homework aur teacher allocation sab subject par chalte hain. EasyEdulab mein 'View Syllabus For This Class' bhi yahin se aata hai.

Kaun use karega: School Admin, Principal

Screens

- Settings > Subjects: naam, code, type (scholastic / co-scholastic), elective flag
- Class-Subject matrix: kis class mein kaunse subject (saal ke hisaab se)
- (Baad mein) Syllabus: chapters aur completion

Features

MUST Subjects master + type + elective flag

MUST Class-subject mapping per academic year, naye saal mein copy-forward

SHOULD Default max marks / pass marks per subject

LATER Syllabus chapters, lesson plan, completion tracking

Data

- `Subject { name, code, type, isElective }` (year-free)
- `ClassSubject { academicYearId, classId, subjectId }` (year-bound)

Edge cases (zaroor socho)

- Language choice (Hindi / Sanskrit / French): student ko alag subject
- Class 11-12 streams (Science / Commerce / Arts) ke alag subjects

Rules

- Subject delete tab tak nahi jab tak marks/timetable mein use ho

Kya chahiye pehle: Academic Year.

Done jab:

- [ ] Subjects add, class se map, naye saal mein copy; isolation test pass

M5 Teacher Allocation & Class Teacher

Roadmap step 5 | MVP | Effort M

EasyEdulab ke section table mein 'Class Teacher' column hai. Kaun teacher kaunsa subject kis class mein padhata hai, usi se attendance, marks aur homework ke access chalte hain.

Kaun use karega: School Admin | Principal | Teacher (apni dekhega)

Screens

- Allocation matrix: teacher x subject x class/section
- Section par 'Class Teacher' chunne ka dropdown (Classes page mein)
- Teacher ka 'My classes' dashboard

Features

MUST Teacher-subject-class allocation per academic year

MUST Section ka class teacher per year (ek section ka ek)

MUST Teacher ko sirf apni classes ka access

SHOULD Workload warning (hafte ke periods)

LATER Auto timetable suggestion

Data

- `TeacherAllocation { academicYearId, teacherId, subjectId, classId, sectionId }`
- `Section.classTeacherId` ki jagah year-wise `ClassTeacher` (kyunki saal badalne par badalta hai)

Edge cases (zaroor socho)

- Teacher beech saal mein chhod gaya: history rahe, naya teacher assign ho (S-3.2)
- Ek teacher ke paas do sections ka class-teacher-ship nahi

Rules

- Class teacher sirf active staff ho sakta hai
- Teacher deactivate hone par allocation reassign ka prompt

Kya chahiye pehle: Staff, Subjects, Academic Year.

Done jab:

- [ ] Allocate, class teacher dikhe, teacher ko sirf apni classes dikhein; tests

M6 Students & Admission

Roadmap step 6 | MVP | Effort XL

Ye school ka dil hai: fees, attendance, exams sab student par. Sabse zaroori design baat: **Student (insaan) aur Enrolment (is saal kis class-section mein) alag rakho.** Isi se 'har cheez Academic Year se jude' wala rule students par lagta hai.

Kaun use karega: School Admin | Principal | Accountant (view) | Teacher (apni class)

Screens

- Students list: search (naam / admission no / mobile), filters (class, section, status), Excel export
- Admission form (stepper): student, guardians, previous school, documents, fee plan
- Student profile tabs: Overview, Family, Documents, Attendance, Fees, Exams, Timeline
- Bulk import (Excel) with validation report
- Promotion tool (year-end), Admission enquiry pipeline (baad mein)

Features

MUST Admission number (configurable scheme, unique, bina audit ke edit nahi), roll number alag

MUST Master: naam, DOB, gender, blood group, address, category / caste (config), religion, mother tongue, photo, previous school

MUST APAAR / PEN field **consent ke saath** (pending / given / refused / withdrawn)

MUST Class + section (ya sirf class, jab section na ho) mein placement, `Enrolment` record per year

MUST Guardians (father / mother / guardian) + primary mobile (SMS isi par jayega); sibling auto-detect

MUST Status: active, TC issued, passed out, left, detained

MUST Year-end bulk promotion, Excel import validation ke saath

SHOULD Duplicate detection (naam + DOB + mobile)

SHOULD Section capacity warning (CBSE norm: section mein optimum 40 tak)

LATER Online admission form (parent khud bhare), enquiry-to-admission funnel, entrance test

Data

- `Student` (insaan, year-free): name, dob, gender, category, photo, ids
- `Enrolment { studentId, academicYearId, classId, sectionId | null, rollNo, status }` (year-bound)
- `Guardian` + `StudentGuardian` link, `ConsentLog`, `Document`

Rules

- Mobile 10 digit (+91), DOB sanity (class ke hisaab se umar), admission no. unique per school
- Class/section delete guard `services/usage.ts` mein ready hai: students aate hi count wahin wire karo. **Bina section wali class ke students class se judte hain** (`sectionId = null`), isliye class-level count bhi chahiye
- Aadhaar optional, masked, role-gated; APAAR admission ki shart nahi (Part 6)

Edge cases (zaroor socho)

- Mid-year admission; transfer-in bina TC ke (RTE: TC ki deri admission rokne ka kaaran nahi, verify karo)
- Teen bachche teen classes mein, ek parent login (S-2.1)
- Duplicate admission merge bina fee/attendance history khoye (S-2.7)
- Naam / DOB ki spelling correction: audit log ke saath
- Guardian custody change: ek parent ka access hatana (S-2.6)
- RTE 25% EWS/DG category student: fee zero, reporting alag

Kya chahiye pehle: Academic Year, Classes (done), parent-identity decision.

Done jab:

- ☐ Admission → profile → class mein placement → list/search/export
- ☐ Section/class delete guards live; promotion dry-run
- ☐ Import 200 rows ka validation report; isolation + year tests

M7 Parents, Family & Consent

Roadmap step 6 | MVP | Effort M

Parent app aur SMS ka base. Ek parent ke kai bachche, ek bachche ke do guardians, aur consent ka record: ye sab yahin set hota hai.

Kaun use karega: School Admin, Parent

Screens

- Student > Family tab
- Parent login (mobile OTP) aur child switcher
- Consent log (kisne, kab, kis cheez ka)

Features

MUST Guardian master: relation, mobile, email, occupation

MUST Ek parent, kai bachche (sibling mapping); ek login se sab dikhein

MUST Consent log: APAAR, photo/media, data sharing; withdraw bhi record ho

SHOULD Guardian ka access revoke (custody case) authorised person se

LATER Dusre branch ke bachche bhi ek hi parent login se

Data

- `Guardian`, `StudentGuardian { relation, isPrimary, canPickup }`
- `ConsentLog { studentId, type, status, givenBy, at }`

Edge cases (zaroor socho)

- Single parent
- Mobile number badal gaya / shared number
- Parent ke bachche do schools mein (Part 8 decision)

Rules

- Parent sirf apne linked bachche dekhe
- Primary mobile verify (OTP) hone par hi SMS 'verified' maane

Kya chahiye pehle: Students.

Done jab:

- [ ] Sibling switcher ka data tayyar, consent log likhna/padhna, tests

M8 Documents Vault

Roadmap step 6 | MVP | Effort M

Birth certificate, Aadhaar copy, previous TC, photo: admission ke saath aate hain aur bahut sensitive hain.

Kaun use karega: School Admin | Principal

Screens

- Student > Documents tab: upload, preview, status (received / pending)
- Admission checklist (kaunse documents zaroori)

Features

MUST PDF / JPG / PNG, size limit, type validation

MUST Encrypted storage (India region), signed short-lived URLs, access log

SHOULD Per-class required-documents checklist aur 'pending' reminder

LATER DigiLocker verification, virus scan queue

Data

- `Document { studentId, type, fileKey, uploadedBy, verifiedBy }` (file object storage mein, DB mein sirf key)

Edge cases (zaroor socho)

- Galat document upload
- Student ka TC issue hone ke baad documents ka retention

Rules

- Public URL kabhi nahi
- Aadhaar copy sirf authorised role ko; har view audit mein

Kya chahiye pehle: Students, file storage service.

Done jab:

- [ ] Upload / view / delete (soft) with audit; unauthorised role ko 403; test

M9 Attendance

Roadmap step 8 | MVP | Effort M

Teacher ki roz ki aadat banane wala module. Chhota hai, roz use hota hai, aur parent ko SMS ka pehla asli fayda yahin milta hai.

Kaun use karega: Teacher | Principal | School Admin | Parent (view)

Screens

- Take attendance: class/section + date; sab default Present, absent par tap, Late / Half-day / Leave
- Today summary (Principal): class-wise %, absent list
- Student attendance calendar (green / red / yellow)
- Holiday calendar

Features

MUST Daily attendance: Present, Absent, Late, Half-day, Excused leave

MUST Cut-off time ke baad lock; edit sirf Admin, reason + audit ke saath

MUST Absent par parent ko notification (messaging se judega)

MUST Monthly / term % report card ke liye auto

MUST Holiday calendar: Sunday, tyohar, exam days attendance mein nahi girte

SHOULD Period-wise attendance (senior classes)

LATER Biometric / RFID / QR scan, staff attendance

Data

- `AttendanceRecord { academicYearId, date, enrolmentId, status, markedBy, markedAt, lateMinutes?, remark? }` unique (enrolment, date)
- `AttendanceSession { classId, sectionId, date, lockedAt }`, `Holiday { date, name }`

Edge cases (zaroor socho)

- Teacher bhool gaya: Admin backfill (S-4.1)
- Galti se absent, SMS ja chuka: correction notice (S-4.2)
- Half-day aur absent-alert (S-4.3); school trip ka 'excused' (S-4.4); lambi medical leave (S-4.5)

Rules

- Future date nahi
- Backfill alag flag se
- Percentage ka denominator admission ki date se (mid-month join)

Kya chahiye pehle: Students (Enrolment), Teacher allocation, Academic Year, Holiday calendar.

Done jab:

- [ ] Ek class 1 minute mein mark hoti hai
- [ ] Lock + audit + % calculation tests
- [ ] Isolation test

M10 Notices & Communication

Roadmap step 9 | MVP | Effort M-L

School aur parent ke beech WhatsApp groups ki jagah audit trail wala channel. Pehle in-app notice, phir SMS / WhatsApp provider jodo.

Kaun use karega: Principal | School Admin | Teacher (apni class) | Parent (read)

Screens

- Notice banao: audience (sab / class / section / role / custom), attachment, schedule
- Notice list + read receipts
- Delivery log (kisko gaya, kisne padha)
- Notification settings (quiet hours)

Features

MUST Notice board, audience segmentation, read receipts

MUST Delivery log (dispute mein kaam aata hai, S-7.4)

SHOULD WhatsApp Business + DLT SMS + push + email, WhatsApp fail par SMS fallback

SHOULD Fee-overdue jaise conditional audience (S-7.5)

LATER Parent-teacher two-way chat (audit trail, non-deletable), polls, IVR

Data

- `Notice`, `NotificationJob`, `DeliveryLog { channel, status, at }`, `Template { dltTemplateId }`

Edge cases (zaroor socho)

- WhatsApp par number nahi hai (S-7.1)
- Raat ko parent ka message, teacher ka SLA (S-7.2)
- Parent ne kaha notice mila hi nahi

Rules

- SMS ke liye **DLT template ID** zaroori; WhatsApp templates Meta se approve
- Unverified mobile par flag; opt-out respect; quiet hours; rate limit

Kya chahiye pehle: Students / Guardians; provider accounts (DLT lead time lambi: abhi shuru karo).

Done jab:

- [ ] Notice sahi audience ko jata hai, read count dikhta hai, log milta hai

M11 Fees (Heads, Collection, Receipts, Dues)

Roadmap step 10 | MVP | Effort XL

School ke liye sabse bada dard aur sabse bada selling point (BRD BO-01, BO-02). Yahan galti ka matlab paisa: isliye is module mein sabse zyada tests aur sabse kam shortcuts.

Kaun use karega: Accountant | School Admin | Principal (view) | Parent (pay)

Screens

- Fee heads (Tuition, Admission, Lab, Transport...), Fee structure (class-wise, installments)
- Student fee account / ledger, Concessions (approval)
- Collect fee (counter): admission no. / naam se dhundo, dues dikhen, Cash / Cheque / UPI / Card / DD
- Receipts list + print / PDF, Cheque register (bounce), Refunds
- Dues / defaulters list + reminders, Daily collection report, Online payment reconcile

Features

MUST Fee heads reusable; class-wise structure per academic year; installments (monthly / quarterly / annual) with due dates

MUST Student ledger: debit (dues) aur credit (payments); **entries kabhi edit / delete nahi**, sirf reversal

MUST Concession / scholarship: flat ya %, head-wise, reason + approval

MUST Offline collection (cash, cheque, DD) aur part payment

MUST Receipt: sequential, non-reusable, per school per financial year, letterhead ke saath PDF

MUST Cheque bounce: credit reverse, due dobara khule, penalty, parent ko notice

MUST Late fee rule (per day / flat / slab) aur per-transaction waiver

MUST Daily collection report (cash / cheque / online alag) aur dues report

SHOULD Online payment (Razorpay / Cashfree / PayU), UPI QR, webhook se reconcile

SHOULD Reminders (due se pehle aur baad), paid hote hi band

SHOULD Refund / withdrawal adjustment, sibling discount rule, advance payment

LATER Combined sibling payment, e-mandate / auto-debit, route-wise transport fee

Data

- `FeeHead`, `FeeStructure { academicYearId, classId, items[{headId, amount, schedule}] }`
- `FeeLedgerEntry { enrolmentId, type: due | payment | concession | reversal | refund, amount, ref }` (append-only)
- `Payment { mode, amount, receiptNo, chequeNo?, gatewayPaymentId? }`, `Receipt`, `Concession`, `RefundRequest`
- **Paisa integer mein (paise)**, kabhi float nahi

Rules

- Receipt number atomic counter se (race condition nahi), financial year (April-March) ke saath reset
- Gateway webhook **idempotent** (same payment id do baar nahi)
- Receipt cancel = reason + reversal entry + naya receipt, purana record rahe
- Time zone IST, rounding ek jagah define

Edge cases (zaroor socho)

- UPI se paisa kata par webhook late / fail: parent ko 'verification pending' dikhao (S-5.1)
- Part payment (S-5.2); cheque bounce jab receipt ban chuka (S-5.3)
- Beech saal mein concession: purane dues ka adjustment (S-5.4); refund (S-5.5)
- Late fee waive (S-5.6); gateway badla (S-5.8)
- RTE / EWS bachcha zero fee; staff ke bachche ka discount; fee mid-year badhi
- Promotion par purana due naye saal mein carry-forward
- Counter par din ka cash close (day-end reconciliation)

Kya chahiye pehle: Students (Enrolment), Academic Year; reminders ke liye Messaging; online ke liye gateway account.

Done jab:

- ☐ Property test: jitne receipts utna ledger credit, daily report se match
- ☐ Receipt PDF, bounce flow, part payment, concession approval chalte hain
- ☐ Isolation + year tests; ek hi receipt number do baar kabhi nahi

M12 Timetable & Substitution

Roadmap step 11 | MVP | Effort L

EasyEdulab ka 'Create Time Table' / 'Add Periods' yahin aata hai. Teacher ke free periods aur substitution isi se nikalte hain.

Kaun use karega: Principal | School Admin | Teacher (apna dekhe) | Parent / Student (view)

Screens

- Period slots setup (timing, break)
- Class timetable grid
- Teacher timetable
- Substitution board (aaj kaun absent, kaun free)

Features

MUST Period slots, breaks; class x day x period grid

MUST Conflict check: teacher ek period mein do jagah nahi (FR-6.1)

MUST Teacher aur class ke alag views

SHOULD Substitution: free teacher ke suggestion, sirf us period ka access

LATER Auto-generate, shifts, room allocation

Data

- `PeriodSlot`, `TimetableEntry { academicYearId, classId, sectionId, day, slotId, subjectId, teacherId, room }`, `Substitution { date, entryId, substituteId }`

Edge cases (zaroor socho)

- Saturday half day
- Primary aur senior ke alag timings
- Lab ke double periods
- Do shifts

Rules

- Teacher-subject-class allocation ke bahar entry nahi
- Teacher leave par automatic substitution prompt

Kya chahiye pehle: Subjects, Teacher allocation, Staff leave (optional).

Done jab:

- [ ] Conflict save par block hota hai
- [ ] Teacher ka 'aaj ka schedule' dikhta hai
- [ ] Tests

M13 Homework & Classwork

Roadmap step 12 | MVP | Effort M

Parent app ko roz kholne ki wajah dene wala feature.

Kaun use karega: Teacher | Parent / Student (view)

Screens

- Homework banao: class/section, subject, due date, attachment, publish
- Homework list (teacher aur student view)

Features

MUST Class/section/subject ke hisaab se create, due date, PDF/image attachment

MUST Publish par parent/student ko notification

SHOULD Student ki submission / 'done' mark

LATER Online submission, grading, plagiarism

Data

- `Homework { academicYearId, classId, sectionId, subjectId, dueDate, attachments[] }`

Edge cases (zaroor socho)

- Mid-term aaya student purana homework dekhe ya nahi (S-6.6)

Rules

- Sirf allocated teacher bana sake
- Attachment size / type limit

Kya chahiye pehle: Subjects, Allocation, Notices.

Done jab:

- [ ] Banta hai, sirf sahi class ko dikhta hai, notification jata hai

M14 Exams, Marks & Report Cards

Roadmap step 13 | MVP | Effort XL

Result school ka sabse sensitive kaam hai. Galat marks = bhaari shikayat. Isliye lock, approval, snapshot aur audit sab chahiye.

Kaun use karega: Teacher (marks) | Principal (approve, publish) | School Admin | Parent / Student (result)

Screens

- Exam setup: Unit Test / Mid / Final, weightage, terms
- Exam schedule (date, time, subject, room)
- Marks entry grid (class x subject), AB / absent code
- Grading scale config (percentage / grade / GPA)
- Report card template (board ke hisaab se), remarks, approval + publish
- Result analytics (class average, toppers, weak subjects)

Features

MUST Exam types with weightage (internal + written), max marks validation

MUST Marks entry; submit ke baad lock; Principal reopen kare to audit log

MUST Absent, medical, exempt ke codes: average mein kaise jayenge, rule tay

MUST Grade / percentage / rank auto-calculation

MUST Report card PDF (attendance % + teacher remarks); publish se pehle Principal approval

MUST Published card ka **snapshot**: baad mein marks badle to purana card na badle, naya version bane

SHOULD Co-scholastic grades (art, PE, discipline), fee dues par withhold ka toggle (S-6.4)

LATER Hall ticket / admit card, question paper blueprint, online exams, board registration

Data

- `Exam { academicYearId, termId, name, weightage }`, `ExamSchedule`
- `Mark { enrolmentId, examId, subjectId, marks, status }`, `GradeScale`

- `ReportCard { enrolmentId, termId, snapshot, status: draft | approved | published, version }`

Rules

- Rounding aur pass criteria ek configurable jagah
- Sirf allocated teacher us subject ke marks bhare
- Published ke baad badlav = naya version + parent ko notice

Edge cases (zaroor socho)

- Marks galat bhar diye lock ke baad (S-6.1); exam mein absent (S-6.2)
- Do teachers ek subject (theory + practical) (S-6.3)
- Published card mein galti, parent ne pakdi (S-6.5)
- Grace marks, re-exam, compartment; mid-term aaye student ke adhoore marks
- Elective subjects ka alag grading

Kya chahiye pehle: Subjects, Students, Attendance (percentage), Academic Year.

Done jab:

- ☐ Ek poori exam cycle: setup → marks → lock → card → approve → publish
- ☐ Snapshot / version test; audit log; isolation test
- ☐ Ek real pilot school ka sample report card format match hota hai

M15 Certificates & ID Cards

Roadmap step 14 | MVP | Effort M

TC aur Bonafide roz ka kaam hai aur galti ka matlab legal dikkat. Serial number kabhi repeat nahi hona chahiye.

Kaun use karega: School Admin | Principal (approve) | Parent (request)

Screens

- Certificate templates (letterhead, signature)
- Generate: TC, Bonafide, Character, Fee-paid certificate
- Issued register (serial, date, kisne)
- ID card batch print (QR)

Features

MUST TC with auto serial (kabhi reuse nahi, saalon ke par bhi), Bonafide with purpose text

MUST Issued register aur re-issue **version** ke saath (S-8.2)

MUST TC se pehle dues / clearance check (policy toggle, S-8.1)

SHOULD Fee-paid certificate: ledger se exact match. FRD mein '80G' likha hai, lekin tuition fee ka deduction aam taur par **80C** mein aata hai: CA se confirm karke naam theek karo

SHOULD QR-coded ID cards, batch aur single; reprint request (S-8.4)

Data

- `CertificateTemplate`, `CertificateIssued { type, serial, studentId, snapshot, version, issuedBy }`

Rules

- Snapshot store karo: baad mein student ka data badle to purana certificate na badle
- TC issue hote hi student status 'TC issued' aur attendance / fee band

Edge cases (zaroor socho)

- TC mein galti: naya serial ya same? (version policy tay karo)
- Visa ke liye urgent bonafide (S-8.3)
- Duplicate TC

Kya chahiye pehle: Students, Fees (dues), Letterhead / logo upload.

Done jab:

☐ TC → status change → register mein entry; serial test; PDF letterhead ke saath

M16 Parent & Student Portal / App

Roadmap step 15 | MVP | Effort XL

Parent ke liye asli value: attendance alert, fee pay, homework, report card ek jagah.

Kaun use karega: Parent | Student

Screens

- Login (mobile OTP)
- Child switcher + dashboard
- Attendance calendar
- Fee due + 1-tap UPI
- Homework, notices, timetable
- Report cards (PDF download)

Features

MUST Sibling switcher, attendance, fee dues, homework, notices, report card

MUST Hindi + English

SHOULD Online fee payment, offline cache (last known data), push notifications

LATER Native app (Flutter) ya PWA, student ka apna login, events / photos gallery

Data

- Naya data nahi: baaki modules ke read-only views + parent session / device record

Edge cases (zaroor socho)

- Kamzor network (S-11.2)
- Do bachche, alag fees, ek payment

Rules

- Parent sirf apne linked bachche
- Device badalne par purane sessions band (S-11.3)
- Feature phone wale parent ke liye SMS + payment link (S-11.1)

Kya chahiye pehle: Students, Attendance, Fees, Notices, Exams (jitne ready, utna dikhao).

Done jab:

- [ ] Ek parent do bachchon ko switch karke dekh sake; tests; real phone par check

M17 Reports, Analytics & UDISE+ Export

Roadmap step 17 | MVP | Effort M-L

Principal aur trust ko number chahiye, aur sarkari form saal ke end mein jaan na nikaale.

Kaun use karega: Principal | School Admin | Accountant

Screens

- Principal dashboard: aaj ki attendance, fee collected, defaulters
- Standard reports (Excel / PDF)
- UDISE+ data export
- Scheduled email report

Features

MUST Class-wise strength, category / gender-wise strength, attendance summary, fee collection, dues, exam analytics

SHOULD UDISE+ export (latest schema ke hisaab se), RTE / EWS report

LATER Custom report builder, scheduled reports, trend charts

Data

- Naya data nahi; heavy reports ke liye aggregation / cache

Edge cases (zaroor socho)

- Saal beech ke admissions / TC ka count kaise
- Class strength jab students ka status badla

Rules

- Report mein tenantId + year filter zaroori (aggregate guard pehle se hai)
- Bade reports background job se

Kya chahiye pehle: Jis module ki report ho woh.

Done jab:

- [ ] Har report ke numbers source screens se match

M18 Staff Directory, Leave & (later) Payroll

Roadmap step 18 | Phase 2 | Effort L

BRD ke Must items (directory) pehle; leave workflow 'Should'. Payroll alag aur bhaari (PF, ESI, TDS), isliye baad mein.

Kaun use karega: School Admin | Principal | Staff

Screens

- Staff profile (qualification, joining, designation)
- Leave types, balance, apply, approve
- Staff attendance

Features

MUST Directory: Teaching / Admin / Transport / Support, search, filter

SHOULD Leave apply, multi-level approval, escalation jab approver available na ho (S-3.4)

SHOULD Aadhaar / PAN masked by default, sirf authorised role ko dikhe

LATER Payroll: salary structure, deductions, payslip; staff attendance biometric

Data

- `StaffProfile`, `LeaveType`, `LeaveRequest`, `LeaveBalance`

Edge cases (zaroor socho)

- Do branch mein padhane wala teacher (S-3.3)
- Resign: account band, history rahe

Rules

- Leave approve hone par substitution ka prompt
- Sensitive field access audit mein

Kya chahiye pehle: Staff, Academic Year.

Done jab:

- ☐ Leave apply → approve → balance update; tests

M19 Phase 2 modules: Transport, Library, Hostel, Assets, Admission CRM, Events

Roadmap step 18 | Phase 2 | Effort M each

BRD ne inhe Phase 1 se bahar rakha hai (Transport live-tracking, Library, Hostel). Pilot ke baad school ki maang dekh kar chuno. EasyEdulab ke menu mein ye sab hain: copy mat karo, zaroorat dekho.

Kaun use karega: Alag-alag roles

Screens

- Transport: routes, stops, vehicles, drivers, bachche per stop, fee link, (baad mein GPS)
- Library: catalogue, issue / return, fine, 'kiske paas kya hai'
- Hostel: rooms, allocation, mess, visitors
- Assets / Inventory: items, location, condition, issue to
- Admission enquiry CRM: enquiry → visit → admission; Events / calendar

Features

LATER Har module ka minimum version pehle: list + assign + simple report

LATER Fee link (transport / hostel) fee engine ke zariye, alag paisa-logic nahi

Data

- Har module ke apne models, sab `tenantScope` + `academicYearId` (jahan lage)

Rules

- Module toggle (SaaS panel) se school-wise on / off

- Naya module bhi isolation test ke bina merge nahi

Edge cases (zaroor socho)

- Transport: stop badla mid-year, fee prorata

Kya chahiye pehle: Students, Fees.

Done jab:

- ☐ Pilot school ki maang confirm hone ke baad hi shuru

M20 SaaS Control Panel (Super Admin)

Roadmap step 19 | MVP (partly done) | Effort L

Aapka business isi se chalta hai. Onboarding, approve aur suspend ho chuke hain; paisa aur nigrani baaki hai.

Kaun use karega: Super Admin (aap)

Screens

- Plans (Basic / Standard / Premium), school ka plan
- Module toggles per school
- Billing / invoices
- Usage analytics, support tickets
- Data export / offboarding
- Support access (kisi school ka data dekhna: consent + audit)

Features

MUST Plan tiers aur feature gating per tier (BR-M01)
MUST Module toggle: band karne par data delete nahi, sirf chhupe (FR-10.3)
SHOULD Billing: flat annual aur per-student/month (BR-M02), grace period + read-only on lapse (S-10.1)
SHOULD Usage analytics, support ticket queue
SHOULD Self-service data export school ke liye (S-10.5)
LATER Trust-level dashboard, white-label, custom SLA

Data

- `Tenant.subscription`, `ModuleToggle`, `Invoice`, `SupportTicket`

Rules

- Super admin ka school-data access audit ke saath aur reason ke saath (S-10.4)
- Plan se jude flags backend mein check ho, sirf UI mein nahi

Edge cases (zaroor socho)

- Mid-year upgrade proration (S-10.3)
- Do schools, ek trust (Part 8 branch decision)

Kya chahiye pehle: Pilot ke baad hi.

Done jab:

- ☐ Plan badalne par features sach mein on / off, tests

PART 5 Kaise banayein aur manage karein

Har module ka recipe (10 step)

1 Reference padho
EasyEdulab / Infinity ke screenshots: list, form, empty state, edge cases. Sirf structure lo; asli student data kabhi repo ya chat mein nahi.

2 Mini-spec likho
Is document ke module playbook se: entities, rules, edge cases. Kya **nahi** banana wo bhi likho.

3 Model
`tenantScope` plugin + (jahan lage) `academicYearId`. Indexes tenantId ke saath.

4 Service
Har function `tenantId` (aur `academicYearId`) pehle argument. Delete / update `Model.deleteOne({ _id, tenantId })` se.

5 Routes + validation
Zod schemas, `requireRole`, errors `ApiError` ke code ke saath.

6 Tests
Happy path, role test, edge cases, **isolation test**, year test, paisa wale module mein property tests.

7 Frontend page
RTK Query, tones / colours, tooltips, empty / loading / error state, phone layout.

8 Real browser check
Throwaway stack (alag ports + alag DB) par desktop aur phone dono; console errors zero.

9 Docs
README mein 'kaise kaam karta hai, file-by-file'; is playbook ka status update.

10 Review
Commit / push tabhi jab aap kahein.

Definition of Done (har module ke liye)

- ☐ Typecheck, lint, build, saare tests pass
- ☐ `<module>.isolation.test.ts` hai aur pass
- ☐ Year-bound hai to `academicYearId` hai aur registry test mein sahi list mein hai
- ☐ Role-wise permission test (kaun kar sakta hai, kaun nahi)
- ☐ Sensitive action par audit log
- ☐ Empty, loading, error, aur 'bahut zyada data' (pagination) states
- ☐ Desktop aur phone par real browser mein dekha; console error zero
- ☐ Tooltips / hints; text honest (jo bana nahi use claim nahi)
- ☐ README aur is playbook ka status update
- ☐ Parked list / decisions mein naya kuch nikla to likha

Testing ke layers

Unit + API tests (vitest)	Har service aur route; real local Mongo, apna test DB.
Isolation tests	Doosre school ko 404 aur data na badle (helper pehle se hai).
Registry tests	Har model scoped ya listed; year-free list explicit.
Money tests	Property-style: random payments ke baad ledger = receipts; duplicate webhook; bounce.
Browser checks	Puppeteer / Edge, throwaway DB, screenshots dekh kar (aapke saath jaisa abhi hua).
Import tests	Ganda Excel: duplicate rows, galat mobile, khaali columns; report saaf aaye.
Pilot UAT	Edge-cases doc (04) ke scenarios school ke saath, 'Confirmed / Needs change' mark karke.

Environments

Local (dev)	Aapka laptop: backend 5000, frontend 5173, docker Mongo. Real data kabhi nahi.
Throwaway (test)	Alag ports + alag DB, check ke baad drop. Aapke dev server ko haath nahi.
Staging	Production jaisa setup, nakli data; pilot school ko demo yahin.
Production	India region, daily backups, monitoring, alerts, secrets env mein.

Pilot school ke saath rollout

1. Pilot school chuno (dost jaisa, 300-800 bachche, ek board). Success ke 3-4 numbers likho (jaise 'fee receipts 100% system se').
2. Pehle unka data lo: students ka Excel, fee structure sheet, report card sample, TC / receipt ka purana format.
3. Staging par unka data import karke unhe dikhao; unke feedback se formats theek karo.
4. Go-live se pehle 2 hafte 'parallel run': system aur purana tareeka saath, roz tulna.
5. Training: accountant (1 din), teachers (1 ghanta attendance + marks), principal (dashboard).
6. Pehle mahine mein roz ek helpline (WhatsApp group), hafte mein ek review call.
7. Pehle exam cycle ke baad report card aur fee ledger ka audit.

Kya nahi karna

- Sab modules ek saath shuru karna (EasyEdulab ke 17 menu items copy karna). Ek module poora, phir agla.
- Paisa, marks ya attendance ko hard-delete ya edit karna. Hamesha reversal + audit.
- Paisa float mein, ya receipt number 'max + 1' se (race condition).
- Role check sirf UI mein; ya teacher ko poori school ka data.
- Year ke bina record banana ('baad mein jod denge').
- Aadhaar / phone ko logs, exports ya screenshots mein chhod dena.
- Policies code mein hard-code karna.
- Bina test ke merge, ya bina real-browser check ke 'ho gaya' kehna.
- Menu mein 'Soon' placeholders (aap pehle hi hata chuke hain).
- Competitor ke screenshots mein asli bachchon ka data repo / chat mein rakhna.

PART 6 India ke rules aur product par asar

Legal advice nahi. Neeche ki jaankari web research (kai secondary sources) se hai. Har card ke "Verify" box mein likha hai kitna pakka hai aur kya confirm karna hai. Product ke liye jo kaam nikalta hai wo "Product mein" mein hai.

1 DPDP Act 2023 aur DPDP Rules (bachchon ka data)

Kya kehta hai: 18 saal se kam ke bachche ke data par **verifiable parental consent** chahiye. Schools ko Rules ke Fourth Schedule se **partial exemption** milta hai, par sirf school ki apni padhai-likhai ya bachchon ki safety se jude kaam ke liye; exemption blanket nahi hai aur school ko documented karna hota hai ki wo kis par tika hai. Bachchon par targeted ads ya nuksaan wala tracking mana hai. Security, retention aur breach notification ke general duties rehte hain.

Product mein: Consent log (kisne, kab, kis kaam ke liye, withdraw), data minimisation (zaroori fields hi), purpose-limited fields, role-based access, encryption at rest, breach runbook, Privacy Notice aur Terms onboarding mein accept karna, retention policy.

Verify: Medium: sources secondary / commentary hain. Notified Gazette text aur Fourth Schedule dekhkar vakeel se confirm karo.

2 APAAR ID

Kya kehta hai: Secondary reports ke hisaab se Supreme Court (July 2026) aur High Court (Orissa, Dec 2025) ne kaha ki consent form mein parent ko **mana karne ka saaf option** ho, aur Aadhaar ke bina padhai nahi roki ja sakti. NCERT ne (April 2026) kaha ki admission ke liye APAAR zaroori nahi (blog report). APAAR data DPDP ke under aata hai aur kisi private party ko share nahi hota.

Product mein: APAAR field ka status: pending / given / refused / withdrawn. Admission **kabhi block nahi**. Data sirf authorised export mein. Parent ko consent form mein 'mana karo' option.

Verify: Low-medium: asli order aur MoE / CBSE guidance khud padho.

3 UDISE+

Kya kehta hai: Har school ka UDISE+ code (onboarding mein pehle se hai); saal mein student-wise aur school-wise data submit hota hai (category, CWSN, class, gender, teachers, infrastructure). Schema har saal badal sakta hai.

Product mein: Student master mein UDISE fields (category / caste, CWSN, mother tongue) config se; export latest schema ke hisaab se; saal ke start mein schema check ka kaam.

Verify: Har saal naya schema dekho.

4 RTE Act 12(1)(c): 25% EWS / DG seats

Kya kehta hai: Private unaided (non-minority) schools ko entry class mein kam se kam 25% seats kamzor varg / disadvantaged group ke bachchon ko deni hoti hain. Sarkar fee ka reimbursement karti hai (state ke kharch ya school ki asli fee, jo kam ho). Process state-wise hai: online module mein entry, signed report, physical verification.

Product mein: Student ki category (EWS / DG), admission seat type, fee zero ya concession, aur reimbursement claim ke liye report. State rules alag: pilot school ke state ke hisaab se.

Verify: Medium: state ke RTE rules aur latest notices dekho.

5 CBSE affiliation norms (agar CBSE school)

Kya kehta hai: Affiliation letters mein section mein optimum **40 bachche tak**, student-teacher ratio lagbhag **30:1**, section-teacher ratio 1:1.5 jaise nirdesh milte hain. Ye schools ke affiliation documents se aaye hain, bye-laws se seedhe nahi.

Product mein: Section capacity warning (field pehle se hai), 'class strength vs capacity' aur 'student-teacher ratio' report.

Verify: Medium: CBSE ke current bye-laws check karo.

6 TC, admission register, scholar register

Kya kehta hai: TC par serial number, kabhi repeat nahi; admission / scholar register aur records ka retention board / state ke hisaab se. RTE ke tahat TC ki deri admission rokne ka kaaran nahi banni chahiye (verify).

Product mein: Certificate module (serial, version, register); retention policy; DPDP 'erasure' aur statutory retention ke beech ka rule tay karo (S-9.2).

Verify: State / board ke rules dekho.

7 SMS (TRAI DLT) aur WhatsApp Business

Kya kehta hai: Transactional SMS ke liye DLT par entity, sender ID (header) aur **har template** register hona zaroori hai. WhatsApp ke templates Meta se approve hote hain aur opt-in chahiye. Dono mein approval ka samay lagta hai.

Product mein: Notification service templates ke saath (template ID store), opt-out, quiet hours, fallback WhatsApp se SMS. **Paperwork abhi shuru karo, code se pehle.**

Verify: Provider (BSP) ke guide dekho.

8 Online payments

Kya kehta hai: RBI-licensed gateway (Razorpay / Cashfree / PayU). Card ka data kabhi apne server par nahi. Webhook se confirmation, settlement ka reconciliation.

Product mein: Idempotent webhook, 'verification pending' state, daily reconcile report, refund flow.

Verify: Gateway ke merchant onboarding rules (KYC, school ka account).

9 Tax: GST aur fee-paid certificate

Kya kehta hai: School ki education services aam taur par GST-exempt hoti hain. Tuition fee par parent ko income-tax deduction aam taur par **section 80C** mein milta hai (donation wala 80G alag cheez hai). FRD ke FR-8.3 mein '80G' likha hai: naam aur text theek karo.

Product mein: Fee-paid certificate ledger se exact; tuition / development fee ka alag dikhana; naam 'Fee Paid / 80C certificate'.

Verify: Pakka: CA se confirm karo.

10 Aadhaar ka istemal

Kya kehta hai: Aadhaar number sirf tab lo jab zaroori ho; padhai Aadhaar par depend nahi kar sakti. Store karo to mask (last 4), encrypt, role-gate.

Product mein: Aadhaar optional field, masked UI, har view audit mein, logs / exports mein nahi.

Verify: Vakeel se review.

11 Security aur data residency

Kya kehta hai: Bachchon ka PII India region mein; TLS, encryption at rest, backups, access logs, admins ke liye MFA, launch se pehle penetration test.

Product mein: Parked list ke items (bcrypt, HttpOnly cookie, MFA, Redis limits) launch se pehle band karo.

Verify: -

12 Aapke aur school ke beech ke documents

Kya kehta hai: Aap data **processor** hain, school **fiduciary**. Terms of Service, Privacy Policy aur Data Processing Agreement chahiye; school ke parents ke liye consent text ka template bhi.

Product mein: Onboarding mein Terms / Privacy accept (gap list mein pehle se); pilot se pehle vakeel se draft.

Verify: Vakeel.

PART 7 School ka saal: mahine-dar-mahine

Software ko school ke saal ke hisaab se chalna hai, isliye ye calendar module planning ka check hai. (Aam CBSE / State board school; apne pilot school ke hisaab se badlo.)

Samay	School kya karta hai	System ko kya chahiye
January - March	Agle saal ke admissions / registration; annual exams; result; promotion faisle; TC requests; fee dues ki last vasooli.	Students (enquiry), Exams, Report cards, Promotion, TC, Dues report
April	Naya session: rollover, fee structure, class-section, subject allocation, timetable, naye admissions, Q1 fees, ID cards.	Academic Year rollover, Fees, Allocation, Timetable, Admissions
May - June	Summer vacation; late admissions; pending dues; staff training; vacation homework.	Notices, Homework, Dues reminders
July	School reopen; unit tests; parent-teacher meeting; Q2 fees; late fee.	Attendance, Exams, Notices, Fees
August - September	Half-yearly exams, results; UDISE+ ki taiyari; board classes ka registration (CBSE: Class 9-12 ke liye).	Exams, Reports, UDISE+ export
October - November	Tyohar chhuttiyan; Q3 fees; pre-board; admission campaign next year ka.	Holiday calendar, Fees, Enquiry CRM
December	Winter exams / pre-board; Q4 fees; annual day / events; data cleaning.	Exams, Events, Reports
Saal bhar	Roz: attendance, homework, notices, fee counter, certificates, leave, substitution, parent queries.	Daily modules

6 poore din / kaam jo end-to-end chalne chahiye

Admission ka din

Enquiry → form → documents → APAAR consent (pending ho sakta hai) → class-section → fee plan → admission fee receipt → ID card → parent ko welcome SMS.

Fee counter ka din

Admission no. / naam → dues + concession dikhe → part / full payment → receipt print → din ke end mein cash close aur daily report.

Result ka din

Marks lock → principal approve → report card publish → parent ko notification → galti ho to naya version aur notice.

Saal-end rollover

Dues carry-forward → promotion (fail / detained alag) → naya saal current → fee structure copy → class teacher / timetable naye saal ke liye.

Bachcha school chhode

Dues clearance (policy) → TC request → principal approve → TC issue → status 'TC issued' → attendance / fee band → records retention mein.

Naya school onboard

Invite / sign-up → onboarding → approve → classes auto-bane → admin staff ko invite karta hai → import → training → go-live.

PART 8 Decisions jo aapko lene hain

Har decision ke saath meri recommendation aur ye ki wo kis cheez ko rokta hai. **D1 aur D2 sabse pehle**: inke bina Students ka data shape pakka nahi hota.

D1 Parent / teacher ki pehchaan: `User.email` poore platform mein unique hai. Do schools wale parent ya teacher ka kya?

- A: Staff ka login email se; **parents / students ka login mobile OTP se, alag** `GuardianAccount`, jo kai schools ke bachchon se jud sake.
- B: Ek `User` + har school mein `Membership` (role) wala model: ek email, kai schools.
- C: Abhi jaisa chhodo (ek email = ek school).

Mera recommendation: A. Staff ke liye ab jaisa theek hai, parents ke liye mobile OTP indian schools mein natural hai. B bada refactor hai.

Rokta hai: Staff invite (Step 5), Students, Parent portal.

D2 Multi-branch trust: ek tenant mein kai branch, ya har branch alag tenant?

- A: **Har campus = ek tenant (school)**; trust-level dashboard baad mein `groupId` se.
- B: Ek tenant, `branchId` har model par (classes, students, fees...).

Mera recommendation: A pilot tak. Isolation bilkul saaf rehta hai aur aaj ka code (tenantScope) seedha chalta hai. B har query aur har index badalta hai. Trust ko ek login chahiye to baad mein 'group' layer.

Rokta hai: Students ka data shape, SaaS billing.

D3 Bina section wali class ke students.

- Decided: student **class se** judta hai (`sectionId = null`).

Mera recommendation: Isliye class delete guard aur counts class-level par bhi.

Rokta hai: Students.

D4 Online fee: paisa kiske account mein?

- A: Har school ke **apne gateway keys** (Razorpay / Cashfree), aap sirf integrate karte ho.
- B: Aapka platform account aur sub-merchants (zyada regulatory zimmedari).

Mera recommendation: A pehle. B baad mein, jab scale ho.

Rokta hai: Online payments.

D5 SMS / WhatsApp provider.

- DLT entity + sender ID + templates register karwana; WhatsApp BSP chunna.

Mera recommendation: Paperwork abhi shuru karo (hafte lag sakte hain), code baad mein.

Rokta hai: Absent alert, fee reminders.

D6 **Parent app.**

- A: Responsive web / PWA pehle.
- B: Native app (Flutter) shuru se.

Mera recommendation: A. Flutter aapke paas hai, par pehle web par sab kuch sahi karo; native pilot ke baad.

Rokta hai: Portal.

D7 **Report card ka format.**

- Pilot school ka asli sample lo (CBSE / State); template engine board ke hisaab se.

Mera recommendation: Bina sample ke format mat banao: har school ka thoda alag hota hai.

Rokta hai: Exams.

D8 **Hindi UI.**

- i18n (translation keys) abhi se, ya baad mein.

Mera recommendation: Students module se naye pages i18n-ready likho; purane pages baad mein.

Rokta hai: Parent portal.

D9 **Pricing aur plans.**

- Flat annual (chhote school) ya per-student per-month (bade); Basic / Standard / Premium mein kya kya.

Mera recommendation: Pilot free; plans SaaS module ke saath tay karo.

Rokta hai: SaaS panel, module toggles.

D10 **Hosting, email, backup.**

- Mongo Atlas (India region) ya self-host; email Gmail se hatake transactional provider (SES / Brevo / Resend); staging environment.

Mera recommendation: Pilot se pehle: India-region Atlas + transactional email + daily backup + ek restore test.

Rokta hai: Pilot.

D11 **Retention aur erasure policy.**

- TC / admission / fee records kitne saal; parent 'data delete karo' kahe to kya.

Mera recommendation: Vakeel / school board ke rules se likho; product mein policy setting.

Rokta hai: Compliance.

D12 **Pilot school aur uski sharten.**

- Ek dost school, ek board, 300-800 bachche; free pilot ke badle feedback aur data.

Mera recommendation: Jald chuno: unka Excel, fee sheet aur report card sample poore design ko sahi rakhega.

Rokta hai: Sab kuch.

PART 9 Go-live checklist aur agle kaam

Pilot go-live se pehle

Product

- ☐ Pilot cut (Part 3) ke saare modules chal rahe hain, real data ke saath
- ☐ Har module ke isolation, role, year tests pass
- ☐ Phone par attendance 1 minute mein; fee receipt print / PDF sahi
- ☐ Report card pilot school ke format se match

Data

- ☐ Students / staff / fee structure ka Excel import saaf (validation report ke saath)
- ☐ Opening balances (purane dues) check kiye
- ☐ Backup chal raha hai aur **restore ek baar kiya gaya**

Security

- ☐ bcrypt native, HttpOnly cookie + CSP, MFA admins ke liye, Redis rate limits, trust proxy
- ☐ Audit log live; Aadhaar masked; logs mein PII nahi
- ☐ Penetration test ya kam se kam checklist-based security review

Legal / Compliance

- ☐ Terms, Privacy Policy, DPA; parents ke liye consent text
- ☐ APAAR consent flow (mana karne ka option)
- ☐ DLT / WhatsApp templates approved

Operations

- ☐ Monitoring + alerts, error tracking, uptime page
- ☐ Support process (kisko call, kitne ghante mein jawab)
- ☐ Staging aur production alag; secrets env mein; deploy ka runbook

Training aur rollout

- ☐ Accountant, teachers, principal ke training sessions
- ☐ 2 hafte parallel run; helpline WhatsApp group
- ☐ Success ke 3-4 numbers roz / hafte dekhe jayenge

Agle kaam (is order mein)

Aaj / is hafte

- ☐ D1 (parent identity) aur D2 (branch model) ka faisla likh do; 30 minute ka kaam hai.
- ☐ Pilot school dhundo (D12) aur unse Excel, fee sheet, report card sample maango.
- ☐ DLT / WhatsApp ka paperwork shuru karo (D5): approval mein samay lagta hai.

Agle 2 hafte

- ☐ Platform base: audit log + change password + logout-all.
- ☐ Academic Year module (aur year-free registry test).

Agle mahine

- ☐ Step 5: Staff invite aur roles.
- ☐ Subjects, Teacher allocation, Classes page mein Class Teacher.
- ☐ EasyEdulab se Teachers aur Students ke screenshots (list, add form, profile) le aao.
- ☐ Students + Enrolment shuru.

Reference ke liye screenshots jo lene hain (har module se pehle)

EasyEdulab / Infinity Medisetu mein se. **Asli student / staff ka naam, phone, Aadhaar blur karo** ya aisi page lo jahan sirf structure dikhe.

Teachers / Staff	List, add form, invite, profile, permission screen
Students	List + filters, admission form (har step), profile tabs, promotion, import
Attendance	Take attendance, report, calendar
Fees	Fee head, structure, collect screen, receipt, dues, cheque, refund, concession
Timetable	Period setup, class grid, teacher view, substitution
Exams	Exam setup, schedule, marks entry, grading scale, report card, hall ticket
Certificates	TC / Bonafide format, ID card, issued register
Notices / Messaging	Compose, audience, delivery report, templates
Parent app	Dashboard, sibling switch, fee pay, attendance, homework
Super admin	Plans, billing, module toggles, support

BRD / FRD mein ye badlav karne chahiye

- **FR-2.3:** 'consent flag = true ke bina save nahi' ko badlo: APAAR / consent ka **status** (pending / given / refused / withdrawn); admission kabhi isse block nahi.
- **FR-8.3:** '80G Tax Benefit certificate' ko 'Fee Paid / 80C certificate' karo (CA se confirm).
- **FR-1.3:** 'Class → Section → Subject' mein section **optional** likho (bina section wali class bhi valid hai).
- **BR-01 / FR-1.1:** multi-branch model D2 ke jawab ke hisaab se likho.
- **BR-02:** Academic Year ka naya rule: har session-bound record year se bandha; year-free list Appendix A.
- **BR-31:** Parent app 'Must' hai; D6 ke hisaab se 'pehle responsive web / PWA, native baad mein' likho.
- **BR-C01:** DPDP ke Fourth Schedule exemption par school ki documented reliance ka section jodo.

APPENDIX A. Kaunsa data saal se bandha hai

Year-free (saal se alag)

Tenant (school), User	Platform / identity
Department, Class, Section	Structure: Class 5 har saal wahi class hai
Subject	Subject master
Student (insaan ka record)	Identity: naam, DOB, ids
Guardian, StudentGuardian	Family relationship
Staff profile	Identity
FeeHead	Master list
CertificateTemplate	Template
Document (file record)	Student ka document
AuditLog	History: saal se alag, hamesha rehta hai

Year-bound (`academicYearId`)

AcademicYear (khud)	Saal ki definition
Enrolment (student is saal kis class-section mein)	`academicYearId`
ClassSubject	Is saal kis class mein kaunsa subject
TeacherAllocation, ClassTeacher	Is saal kaun padhata hai
TimetableEntry, Substitution	Is saal ka timetable
AttendanceRecord, Holiday	Is saal ki attendance / calendar
FeeStructure, FeeLedgerEntry, Payment, Receipt, Concession	Is saal ke dues aur payments (receipt FY ke saath)
Exam, Mark, ReportCard	Is saal ke exams aur results
Homework, Notice (jahan session se bandha)	Is saal ke
CertificateIssued	Issue ka saal + snapshot

> Naya model banate waqt sawal poochho: 'agar naya saal shuru ho jaye to kya ye record naye saal ke liye dobara banna chahiye?' Haan = year-bound. Naa = year-free.

B. Glossary

APAAR	Automated Permanent Academic Account Registry: 'One Nation One Student ID' (12 digit).
PEN	Permanent Education Number, student ka school-level ID.
UDISE+	Unified District Information System for Education Plus: school aur student ka sarkari data portal.
RTE / EWS / DG	Right to Education Act; Economically Weaker Section; Disadvantaged Group (25% seats).
TC	Transfer Certificate. Bonafide: 'ye bachcha is school ka student hai' ka certificate.
SR / Scholar / Admission no.	School ke registers ke numbers; school ke hisaab se alag naam.
DLT	Distributed Ledger Technology registry (TRAI): SMS ke liye entity, header aur template registration.
DPDP	Digital Personal Data Protection Act 2023 (aur Rules).

Fiduciary / Processor	DPDP mein: data ka maalik / uski taraf se kaam karne wala. School = fiduciary, EduShiksha = processor.
CCE / GPA / Grade	CBSE ke grading tareeke (continuous evaluation, grade point).
Co-scholastic	Padhai ke alawa: art, PE, discipline, skills.
Enrolment	Student ka 'is saal, is class-section mein' record.
Tenant	Ek school (ya campus) ka alag data ka dayra.
Rollover	Naya saal shuru karte waqt purane saal se naya banana: promotion, copy, carry-forward.
Idempotent	Wahi request do baar aaye to asar ek hi baar (payment webhook ke liye zaroori).
Ledger	Khata: har due aur payment ki entry, jo kabhi mitati nahi.

C. Sources (web research)

Ye pages web research se mile; kai secondary hain (blog, commentary, school-hosted copies). Kisi bhi rule par kaam karne se pehle asli sarkari text aur vakeel / CA se confirm karo.

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